

SRM Institute of Science and Technology
College of Engineering and Technology
Common to ALL M.Tech Branches

SRM Nagar, Kattankulathur – 603203, Chengalpattu District, Tamilnadu
Academic Year: 2025-2026 (ODD)

Test: FJ 2

Date: 15.11.2025

Course Code & Title: 21IPC501J Research Methodology

Duration: 100 Min

Year & Sem: I / I

Max. Marks: 60

	Part A (Answer any 3 Questions)	Marks	CO	BL																																																																		
Q 1.i)	A research team repeatedly releases multiple papers that present nearly identical findings to inflate their total number of publications. Discuss the ethical concerns linked to duplicate submissions and breaking a single study into many minimally different papers (“salami publication”). Explain how choosing to report only certain parts of data harms the trustworthiness of scientific work in India. Also, outline measures that academic journals can adopt to detect and prevent these practices.	15	3	3																																																																		
Q 1.ii)	Upholding ethical standards in research has become more essential today than ever before. Justify the statement.	5	3	3																																																																		
Q 2i) .	Three control factors A, B and C each at 3 levels are studied using an L9 orthogonal array. Each run is repeated 3 times. The measured response is a "larger is better" quality characteristic (higher = better). The data: <table><tr><td>Run</td><td>A</td><td>B</td><td>C</td><td>Replicates (y)</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>65, 67, 66</td></tr><tr><td>2</td><td>1</td><td>2</td><td>2</td><td>70, 72, 71</td></tr><tr><td>3</td><td>1</td><td>3</td><td>3</td><td>75, 74, 76</td></tr><tr><td>4</td><td>2</td><td>1</td><td>2</td><td>68, 69, 70</td></tr><tr><td>5</td><td>2</td><td>2</td><td>3</td><td>73, 75, 74</td></tr><tr><td>6</td><td>2</td><td>3</td><td>1</td><td>64, 63, 65</td></tr><tr><td>7</td><td>3</td><td>1</td><td>3</td><td>78, 77, 79</td></tr><tr><td>8</td><td>3</td><td>2</td><td>1</td><td>66, 68, 67</td></tr><tr><td>9</td><td>3</td><td>3</td><td>2</td><td>71, 70, 72</td></tr></table> The results are Grand mean S/N = 36.9528 dB. Factor-level S/N <table><tr><td>Factor</td><td>Level 1</td><td>Level 2</td><td>Level 3</td></tr><tr><td>A</td><td>36.9707</td><td>36.7599</td><td>37.1278</td></tr><tr><td>B</td><td>37.0015</td><td>36.9754</td><td>36.8815</td></tr><tr><td>C</td><td>36.3433</td><td>36.9407</td><td>37.5744</td></tr></table> ANOVA (on S/N): C is the dominant factor with ~90.58% contribution; A ~ 8.15%; B ~ 0.95%. Predicted Performance at Optimal Settings Predicted S/N ~ 37.7981 dB Predicted response (y) ~ 77.6	Run	A	B	C	Replicates (y)	1	1	1	1	65, 67, 66	2	1	2	2	70, 72, 71	3	1	3	3	75, 74, 76	4	2	1	2	68, 69, 70	5	2	2	3	73, 75, 74	6	2	3	1	64, 63, 65	7	3	1	3	78, 77, 79	8	3	2	1	66, 68, 67	9	3	3	2	71, 70, 72	Factor	Level 1	Level 2	Level 3	A	36.9707	36.7599	37.1278	B	37.0015	36.9754	36.8815	C	36.3433	36.9407	37.5744	12	4	4
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	Observed at that setting (Run 7) = 78 - Based on the ANOVA results, which factor has the greatest influence on the performance, and what does its high percentage contribution indicate about the process? - Explain why the combination A3–B1–C3 is considered optimal. What characteristics of the factor-level S/N means lead to this conclusion? - The predicted response at the optimal settings is 77.6, while the observed value is 78. What does this close agreement indicate about the adequacy and reliability of the Taguchi model used? - Factor B shows less than 1% contribution. In practical terms, how should this factor be handled during process control or parameter selection? Should effort be spent on optimizing it? Why or why not?			
Q 2 ii)	Taguchi's robust design methodology is widely used in engineering and manufacturing. Analyse the circumstances under which the Taguchi method is preferred over traditional experimental approaches.	8	4	3
Q 3	A researcher wants to investigate whether different types of practice influence how well rats retain a maze they previously learned. After completing the same two-week training on an initial maze, 18 rats are randomly assigned to one of three treatment groups for an additional one-week period: Group 1 – Same Maze Practice: Rats continue running the same original maze twice per day. Group 2 – New Maze Practice: Rats practice a different maze twice per day for the entire week. Group 3 – No Practice: Rats receive no practice and remain in their home cages. At the end of the week, all rats are tested once on the original maze, and their completion times (in seconds) are recorded: Group 1: 10, 12, 18, 15, 16, 12 Group 2: 25, 22, 27, 30, 28, 25 Group 3: 15, 17, 20, 21, 19, 18 Using a one-way ANOVA, determine whether the treatment groups differ significantly in their performance on the original maze. Clearly state: - The null and alternative hypotheses - The ANOVA table (SSA, SSW, SST, df, MS, F) - The decision at $\alpha = 0.05$ - A brief interpretation of what these results mean for learning and memory retention in rats.	20	5	3
Q 4 i)	A CNC machine produces an average of 2 defective components per hour . What is the probability that in a 2-hour period, the machine produces exactly 3 defective components?	5	5	4
Q 4ii)	A manufacturing company plans to redesign the layout of machines on its production floor to reduce material-handling time and improve workflow efficiency. The layout problem is complex, nonlinear, and combinatorial, with multiple local optima—similar to challenges seen in large-scale scheduling problems such as public-bus route planning. Analyse how global optimization techniques such as Genetic Algorithms (GA) and Particle Swarm Optimization (PSO) can be applied to solve such complex	15	4	4

	layout or scheduling problems. Evaluate their suitability for exploring large search spaces, avoiding local optima, and identifying a robust global solution.			
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